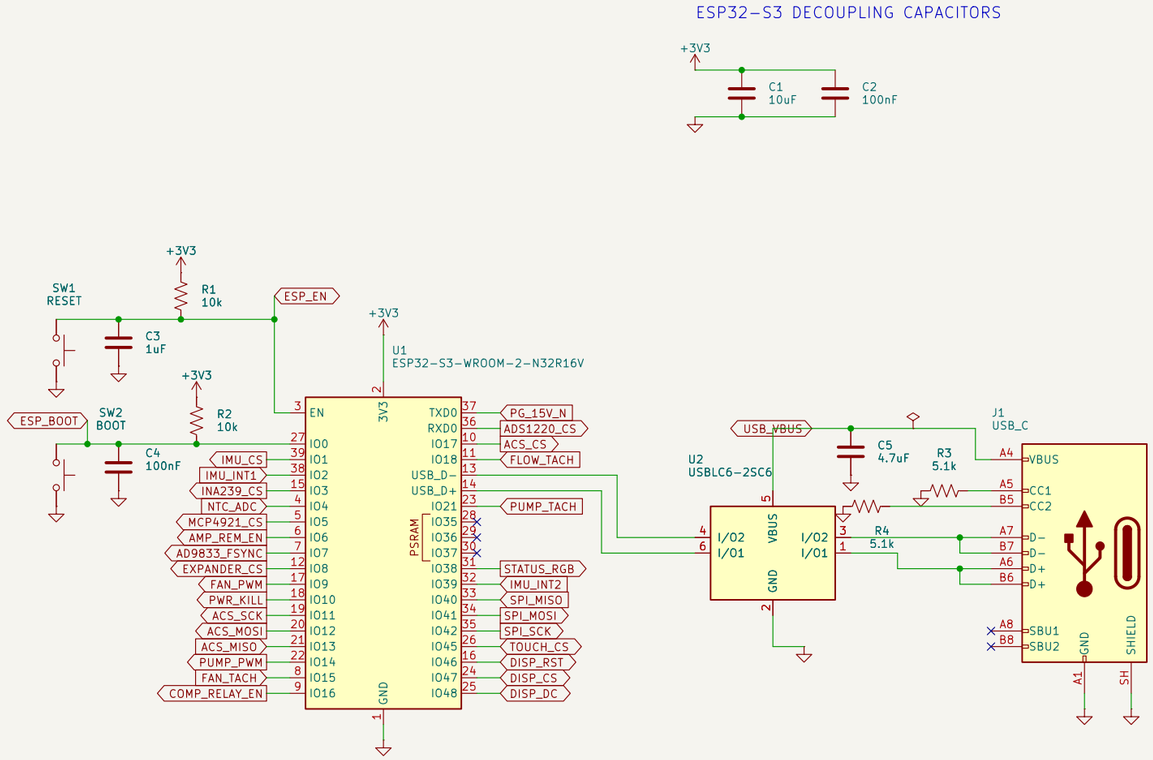
ESP32-S3-WROOM-2-N32R16V: 32 MB octal flash + 16 MB octal PSRAM (GPIO26-37 reserved by module)

GPIO map follows include/config/pin_config.h



Power Input
File: sheets/power_input.kicad_sch

Signal Generator
File: sheets/signal_gen.kicad_sch

Water Cooling
File: sheets/cooling.kicad_sch

Power Button
File: sheets/power_button.kicad_sch

Amp Control
File: sheets/amp_control.kicad_sch

Cold Head Temp
File: sheets/cold_head.kicad_sch

Rails 5V 3V3
File: sheets/rails_logic.kicad_sch

Amp Current
File: sheets/amp_current.kicad_sch

IMU
File: sheets/imu.kicad_sch

Rails +-15V
File: sheets/rails_15v.kicad_sch

Power Good
File: sheets/power_good.kicad_sch

System Current
File: sheets/system_current.kicad_sch

Indicators
File: sheets/indicators.kicad_sch

Compressor Relay
File: sheets/compressor_relay.kicad_sch

GPIO Expander
File: sheets/gpio_expander.kicad_sch

Debug
File: sheets/debug.kicad_sch

Supply Switches

Display

Cryocooler Controller

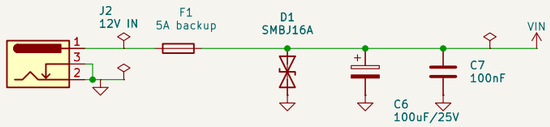
Sheet: /
File: cryocooler.kicad_sch

Title: Cryocooler Controller

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/19

12 V input from a Dell XYK93 (DPS-2000EB class, ~12.2 V, ~160 A available): use the main 12 V rail with PS_ON held low.

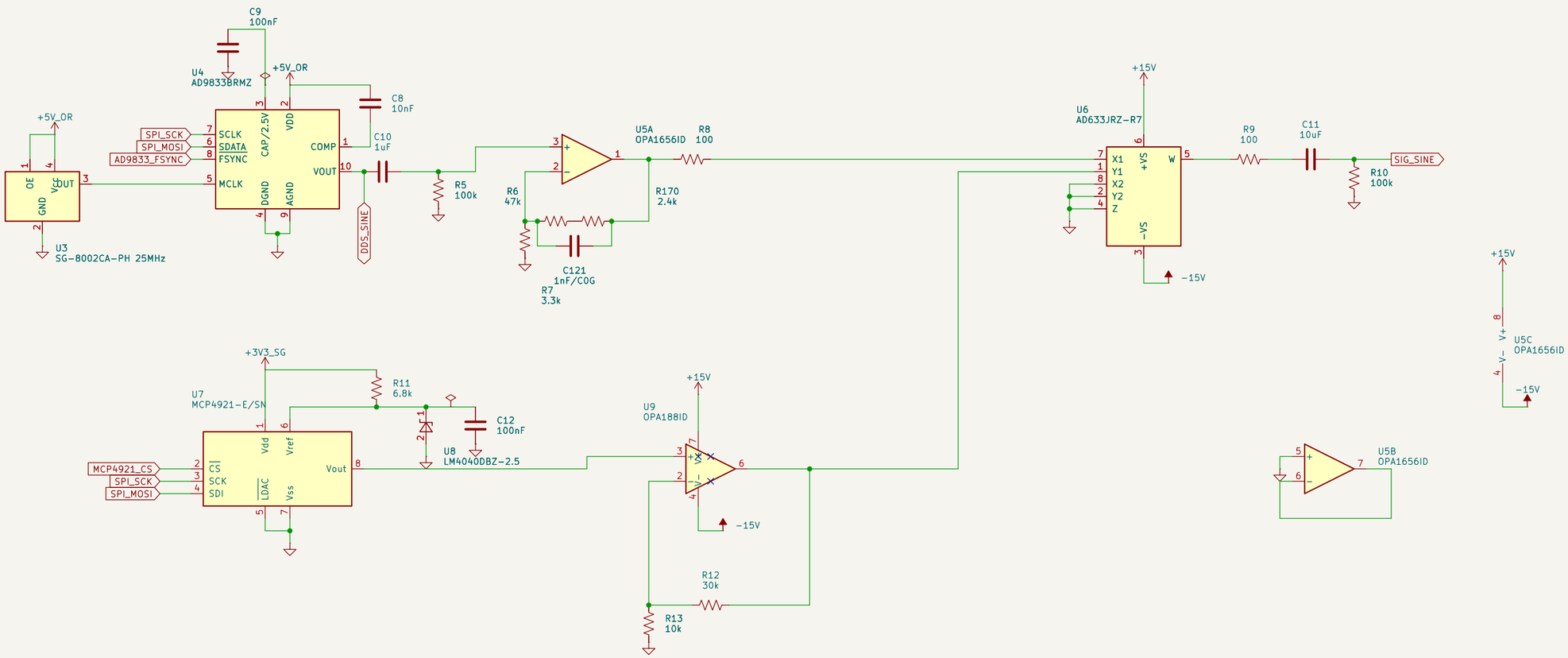
F1 is a backup only: add an inline high-breaking-capacity fuse (7.5-10 A) at the PSU end. Reverse polarity: TVS forward-clamps and blows the fuse.



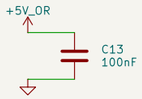
Cryocooler Controller		
Sheet: /Power Input/ File: power_input.kicad_sch		
Title: Power Input		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/19

AD9833 60 Hz sine -> OPA1656 gain stage -> AD633 multiplier; MCP4921 + OPA188 give a 0–10 V amplitude control.

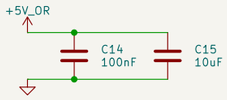
Vctl 0–10 V (2.5 V ref x4) x sine +/−2.3 V pk / 10 = 0..2.3 V pk (4.6 Vpp, 1.6 Vrms) to amplifier input; ZE500.1 RCA input range 0.2–6 V.



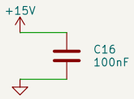
SG-8002CA DECOUPLING CAPACITORS



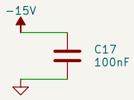
AD9833 DECOUPLING CAPACITORS



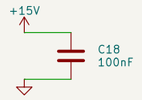
OPA1656 +15V DECOUPLING CAPACITORS



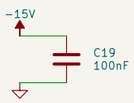
OPA1656 −15V DECOUPLING CAPACITORS



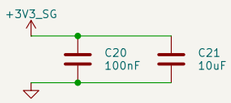
AD633 +15V DECOUPLING CAPACITORS



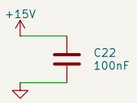
AD633 −15V DECOUPLING CAPACITORS



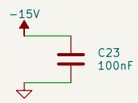
MCP4921 DECOUPLING CAPACITORS



OPA188 +15V DECOUPLING CAPACITORS



OPA188 −15V DECOUPLING CAPACITORS



Cryocooler Controller

Sheet: /Signal Generator/
File: signal_gen.kicad_sch

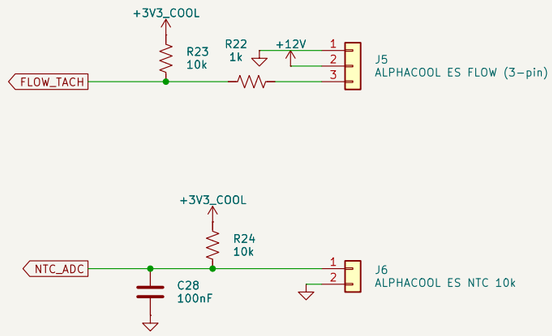
Title: Signal Generator

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/19

Fan and pump: LEDC PWM at 25 kHz through a BSS138, because the PWM input can sit at 5.25 V (Intel 4-wire spec) and GPIOs are not 5 V tolerant. GPIO high = FET on = 0% drive; firmware inverts the duty.

Tach: open collector, 2 pulses/rev, counted with PCNT. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.

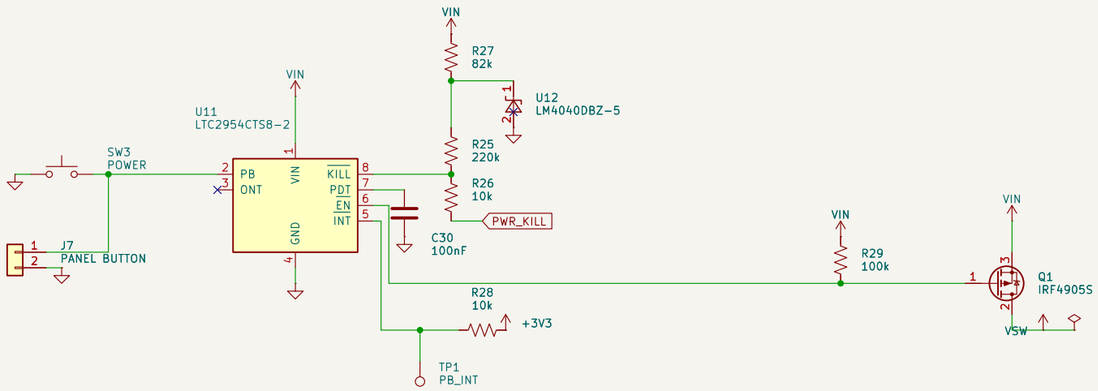




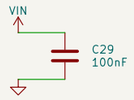
Cryocooler Controller		
Sheet: /Water Cooling/		
File: cooling.kicad_sch		
Title: Water Cooling		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/19

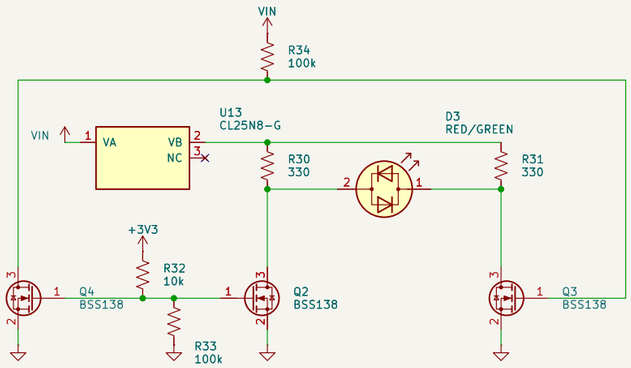
LTC2954-2: tap PB = on, hold = off. EN# drives the P-FET gate; KILL# from GPIO10 (active low, 100 ms pulse).

PWR_ON follows the +3V3 rail: on = green, standby = red. CL25 (25 mA) feeds both legs, ~9 mA reaches the LED, ~25 mA total from VIN in standby



LTC2954 DECOUPLING CAPACITORS

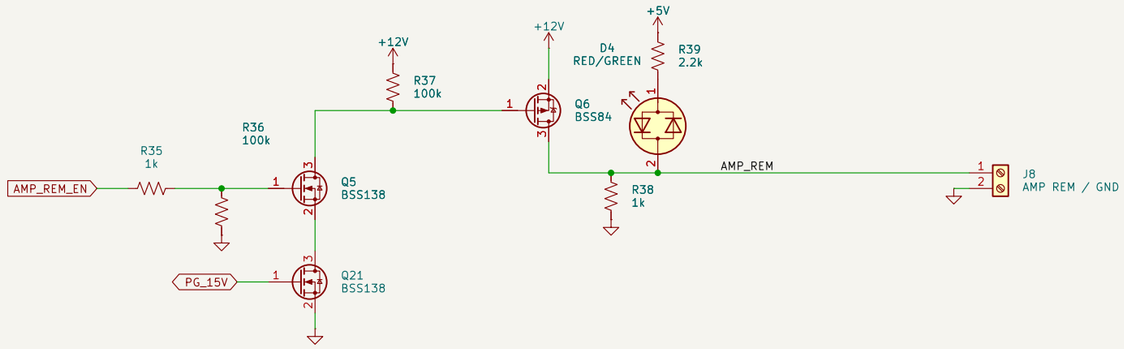




Cryocooler Controller		
Sheet: /Power Button/		
File: power_button.kicad_sch		
Title: Power Button		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/19

GPIO6 high -> +12 V on the amplifier REM terminal (P-FET high-side switch), but only while PG_15V is high (both +-15 V rails good: Q9 in series with the level shifter). LED: green = on, red = off.

Amplifier: AUDIOZERONE ZE500.1 (external): REM terminal + RCA/line input. Its power leads do not pass through this board.





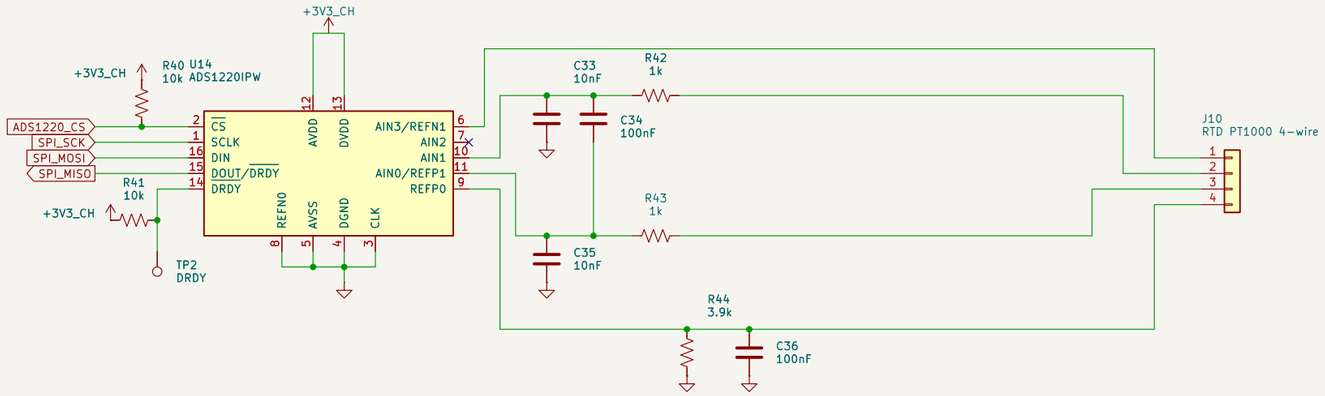
Cryocooler Controller

Sheet: /Amp Control/
File: amp_control.kicad_sch

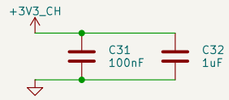
Title: Amp Control

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/19

PT1000 4-wire ratiometric: IDAC1 (AIN3) excites, AIN1/AIN0 sense, Rref (3.9k) on REFP0/REFN0. SPI mode 1 on the main bus (CS GPIO44), internal oscillator (CLK tied low). DRDY is not wired to the MCU.



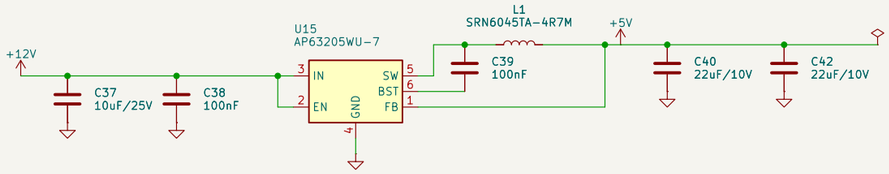
ADS1220 DECOUPLING CAPACITORS

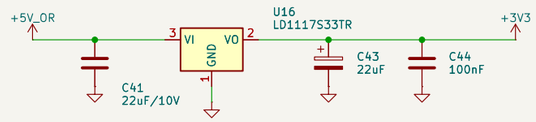


Cryocooler Controller		
Sheet: /Cold Head Temp/ File: cold_head.kicad_sch		
Title: Cold Head Temp		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/19

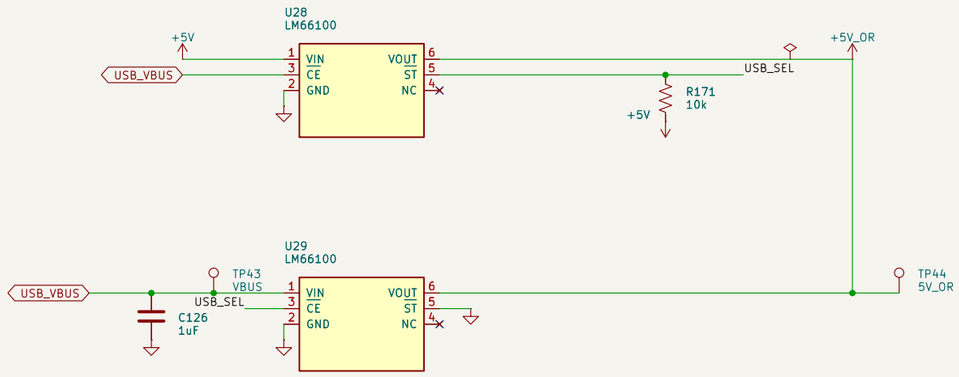
12 V -> 5 V (AP63205 buck, datasheet Fig. 1 / Table 3: 4.7 uH, 10 uF in, 2 x 22 uF out, 100 nF bootstrap) -> 3.3 V (LD1117S33).

AP63205: VIN 3.8–32 V (40 V for 400 ms), EN tied to VIN, FB tied to the fixed 5 V output.





+5V_OR: USB alone powers +3V3 and the DDS, not the +5V rail or the +−15 V stage. LM66100 SLVSEZ8A Fig. 17 (CE to the other input, ST + pull-up to the second CE).



Cryocooler Controller

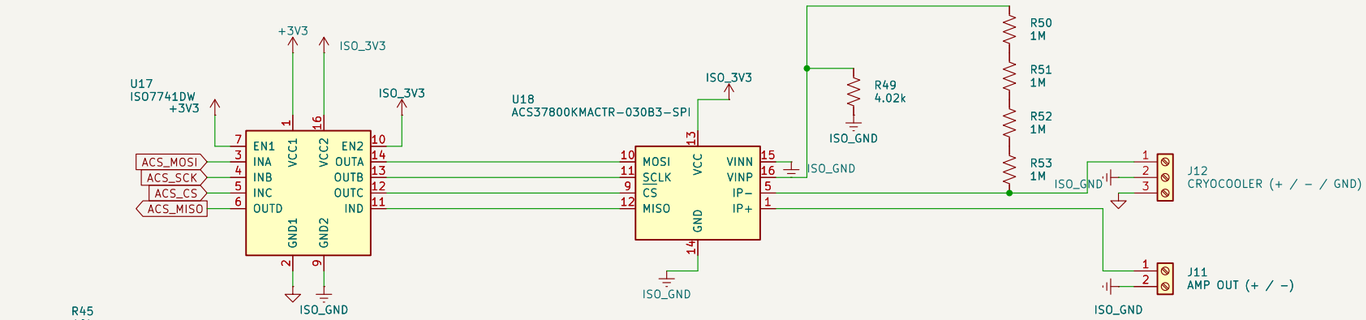
Sheet: /Rails 5V 3V3/
File: rails_logic.kicad_sch

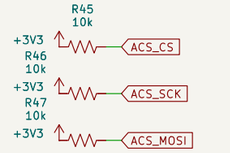
Title: Rails 5V 3V3

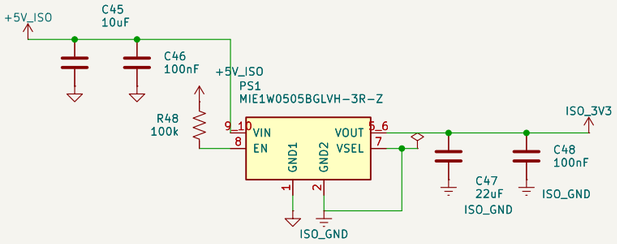
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/19

Amplifier output V/I monitor: ACS37800 (SPI) on an isolated 3.3 V domain (MIE1W0505, VSEL=GND2 -> 3.3 V); SPI bus GPIO11/12/13, CS GPIO17.

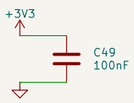
Mains-class creepage/clearance applies between ISO_GND domain and the logic side on the PCB.



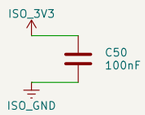




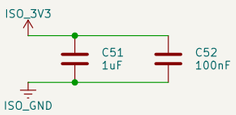
ISO7741 VCC1 DECOUPLING CAPACITORS



ISO7741 VCC2 DECOUPLING CAPACITORS



ACS37800 DECOUPLING CAPACITORS



Cryocooler Controller

Sheet: /Amp Current/

File: amp_current.kicad_sch

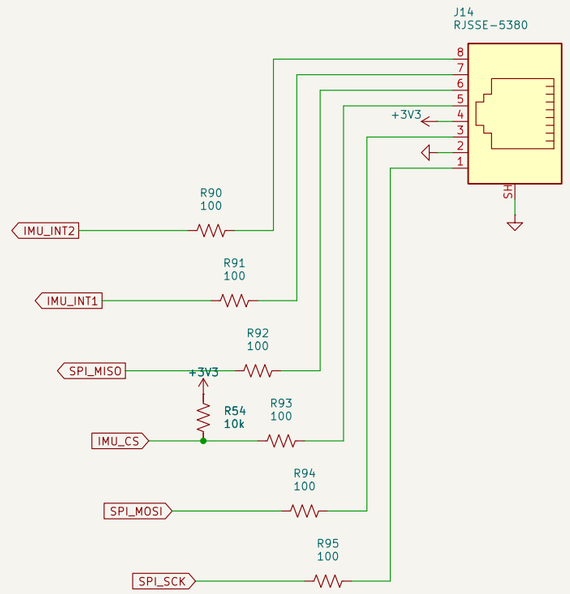
Title: Amp Current

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 9/19

LSM6DSOX is the off-board Adafruit module (product 4438) on the cooler, cabled to J14 (RJ45, straight Cat5e cable) over the main SPI bus (CS GPIO1, INT1 GPIO2 single-tap, INT2 GPIO39 wake-up).

RJ45 pin -> module pad: 1 SCL, 2 GND, 3 SDA, 4 Vin, 5 CS, 6 DO, 7 I1, 8 I2 (3Vo pad open). Pairs: SCK with GND, MOSI with MISO, +3V3 with CS, INT1 with INT2. Plain jack, no magnetics.

Module Vin = +3V3 keeps its logic at 3.3 V. 100 R series resistors damp the cable; the 10 k keeps CS high with the cable unplugged; the jack shield is tied to GND.



Cryocooler Controller

Sheet: /IMU/

File: imu.kicad_sch

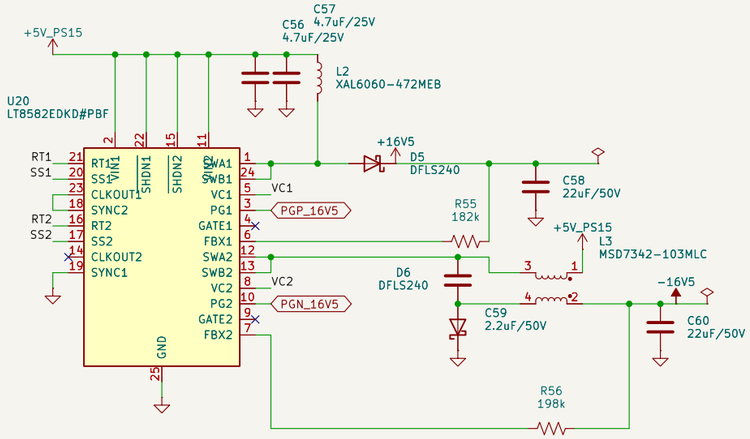
Title: IMU

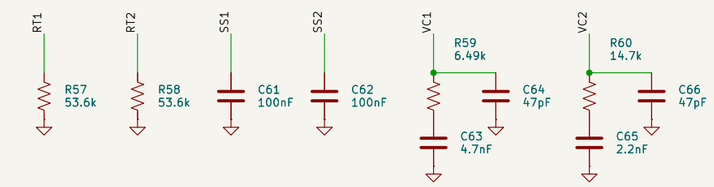
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 10/19

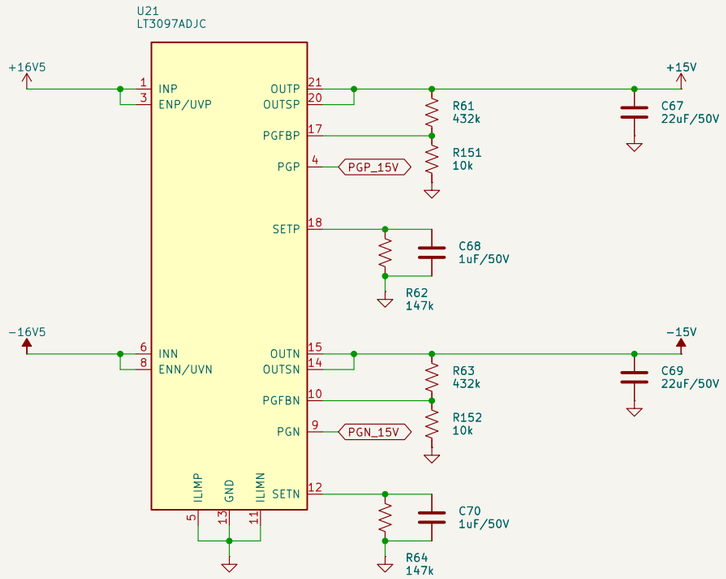
LT8582 (5 V in, 1.5 MHz): channel 1 boosts to +16.4 V, channel 2 inverts to −16.5 V; one LT3097 post-regulates them to +/−14.7 V.

LT8582 circuit after datasheet p.30 (5 V to +/−12 V): RT 53.6k, SS 0.1 uF, VC 6.49k+4.7nF / 14.7k+2.2nF with 47 pF. RFBX1 = (16.4 − 1.204) / 83.3 uA = 182k, RFBX2 = (16.5 + 0.007) / 83.3 uA = 198k. Inverting needs L >= 5.6 uH (duty ~0.80): 10 uH coupled MSD7342.

PG1 / PG2 (open drain, high = within 4 %) and the LT3097 PGP / PGN go to the power_good sheet; GATE1/2 unused (no output disconnect).







Cryocooler Controller

Sheet: /Rails +−15V/
File: rails_15v.kicad_sch

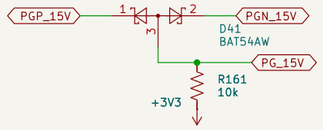
Title: Rails +−15V

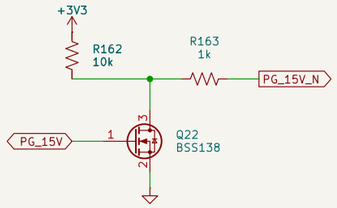
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 11/19

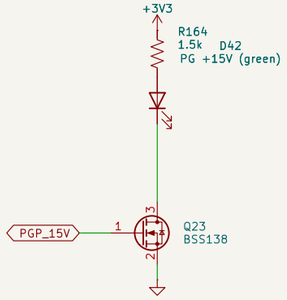
PGP and PGN pull their own node low when bad. Each node is the +15 V rail through 220k/62k (about 3.2 V when good, 0 V if its pin pulls low or the rail is missing) and lights its green LED.

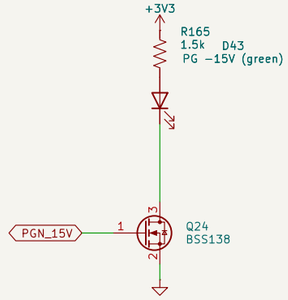
PG_15V = the two nodes ANDed by a BAT54AW with a 10k pull-up to +3V3 (3.3 V only when both are good); it gates the amplifier REM switch and reads LOW on GPIO43 (PG_15V_N) when both rails are good.



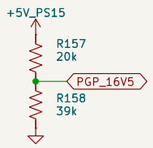


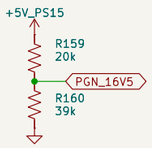






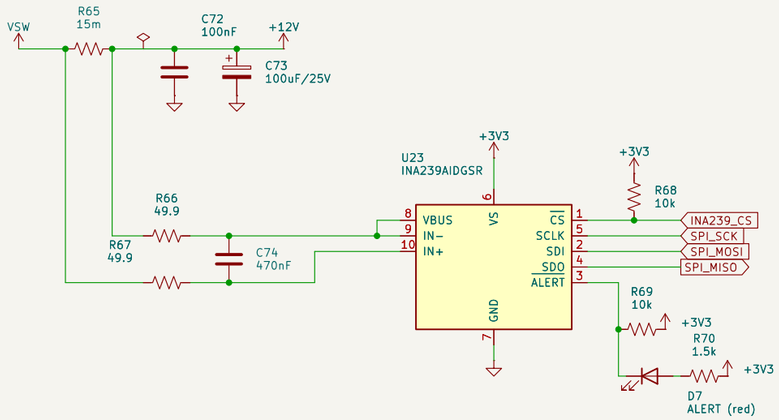
LT8582 POWER GOOD (to MCP23S08 GP2 / GP3)





Cryocooler Controller		
Sheet: /Power Good/		
File: power_good.kicad_sch		
Title: Power Good		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 12/19

INA239 monitors the switched 12 V rail: 15 mOhm shunt, SPI mode 1 on the main bus (CS GPIO3), +/−163.84 mV range (~10.9 A). ALERT only drives its LED.



3V3 POWER LED

INA239 DECOUPLING CAPACITORS

Cryocooler Controller

Sheet: /System Current/
File: system_current.kicad_sch

Title: System Current

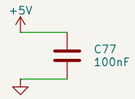
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 13/19

READY (MCP23S08 GP5) and FAULT (GP6) LEDs, active high; WS2812B status LED on GPIO38 via a 74AHCT1G125 3.3 V -> 5 V level shifter (WS2812B VIH = 0.7 x VDD).

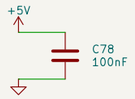


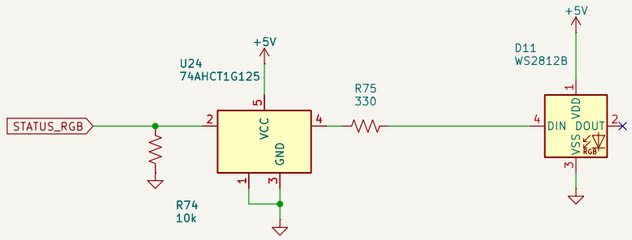


74AHCT1G125 DECOUPLING CAPACITORS



WS2812B DECOUPLING CAPACITORS





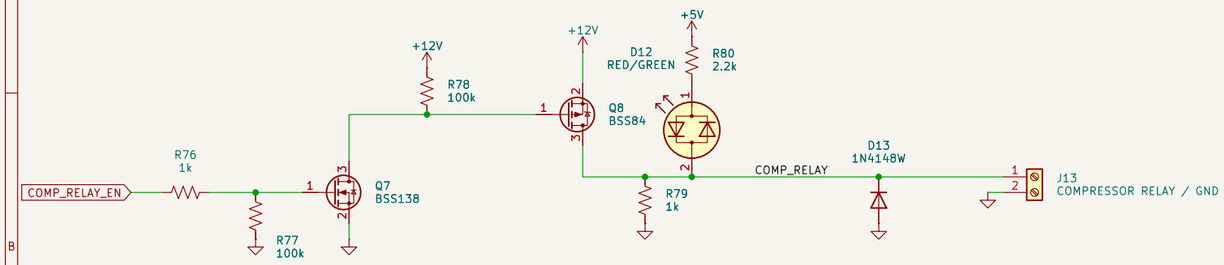
Cryocooler Controller

Sheet: /Indicators/
File: indicators.kicad_sch

Title: Indicators

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 14/19

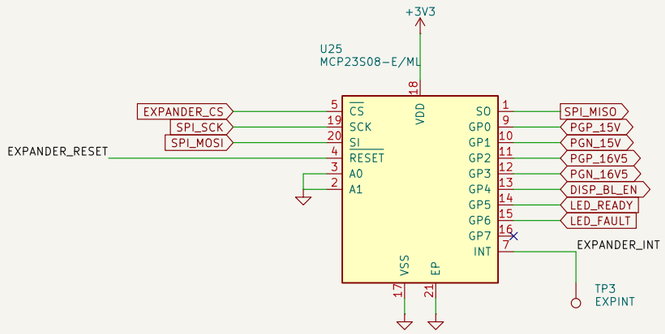
COMP_RELAY_EN (GPIO16) high -> +12 V on the compressor relay terminal (BSS84 high-side switch, 130 mA max: drive an external relay or SSR coil). Flyback diode across the output. LED: green = on, red = off.



Cryocooler Controller		
Sheet: /Compressor Relay/ File: compressor_relay.kicad_sch		
Title: Compressor Relay		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 15/19

MCP23S08: SPI mode 0 slave on the main bus (CS GPIO8). SO releases within 100 ns of CS rising (datasheet Fig. 2–6, parameter 14), so it shares SPI_MISO without a buffer. A1/A0 low: opcodes 0x40 / 0x41.

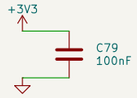
All pins power up as inputs. RESET, A1 and A0 must be biased (datasheet Table 1–1). Exposed pad to VSS, in addition to pin 17. GP7 is unused: firmware enables its weak pull-up.



PULL-UPS



MCP23S08 DECOUPLING CAPACITOR



Cryocooler Controller

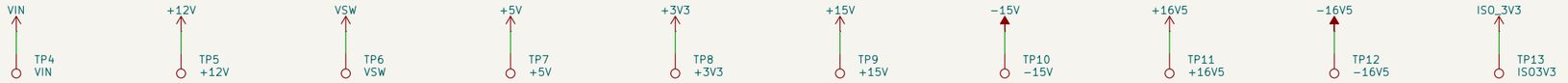
Sheet: /GPIO Expander/
File: gpio_expander.kicad_sch

Title: GPIO Expander

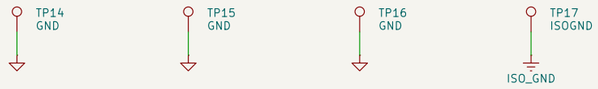
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 16/19

Debug aids: test points (pad per net, loops for ground) and rail LEDs. Each test point sits on a global label or power symbol; the PCB generator places it beside a pad on the same net.

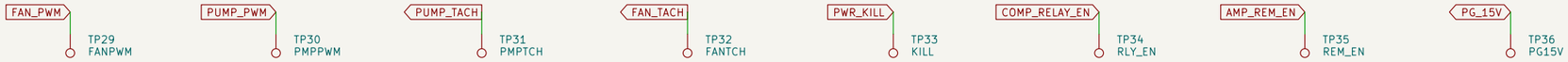
POWER RAILS



GROUND (scope loops)



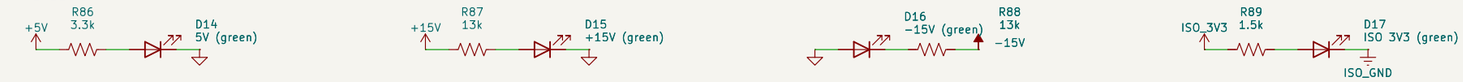
FAN, PUMP, INTERRUPTS AND CONTROL



ANALOG AND SENSORS



RAIL LEDS



Cryocooler Controller

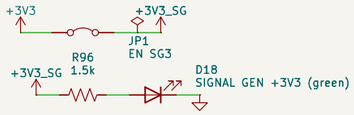
Sheet: /Debug/
File: debug.kicad_sch

Title: Debug

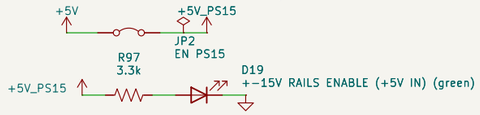
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 17/19

Each Keystone 5110 0603 jumper sits in series with one load supply: remove it to power that part off. The LED shows the switched rail is present. A part left unpowered can still be fed through its signal pins, so disconnect the cable too when you need it fully dead.

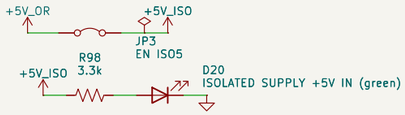
SIGNAL GEN +3V3



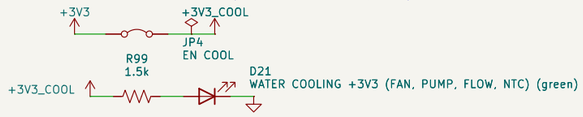
+−15V RAILS ENABLE (+5V IN)



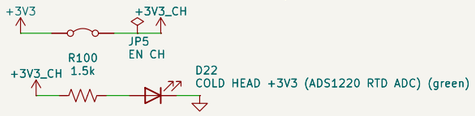
ISOLATED SUPPLY +5V IN



WATER COOLING +3V3 (FAN, PUMP, FLOW, NTC)



COLD HEAD +3V3 (ADS1220 RTD ADC)



Cryocooler Controller

Sheet: /Supply Switches/
File: supply_switches.kicad_sch

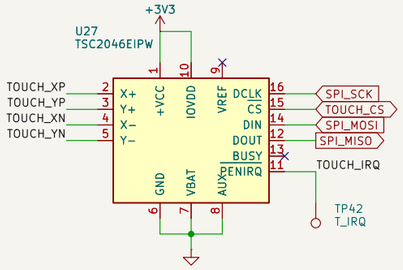
Title: Supply Switches

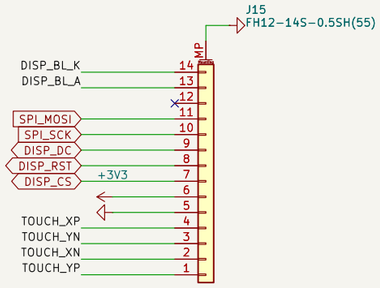
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 18/19

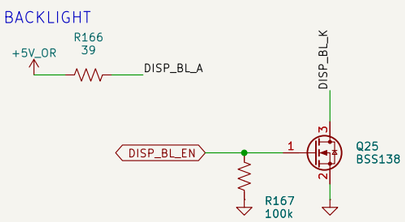
Phoenix DCT T 2,4 QVGA S RTOUCH (in the UCS 125-87-F-GD-D2,4 lid), datasheet 109921: ST7789V, 4-wire SPI half duplex (SDA is input only), VCC 2.6-3.3 V, FPC FH12-14S-0.5SH.

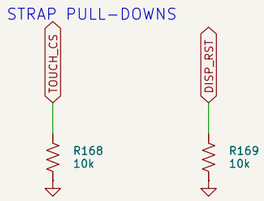
Pin 1 YU, 2 XL, 3 YD, 4 XR (raw resistive touch), 5 GND, 6 VCC, 7 /CS, 8 /RST, 9 RS (D/C), 10 SCL, 11 SDA, 12 NC, 13 A, 14 C. Backlight: 4 LEDs in parallel, 3.0 V / 60 mA typ.

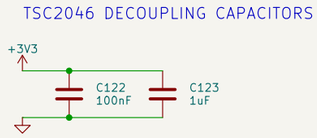
GPIO45 (TOUCH_CS) and GPIO46 (DISP_RST) are strapping pins: 10k pull-downs hold their default (low) levels at reset; firmware drives TOUCH_CS high before any SPI traffic.

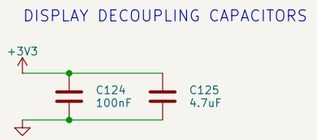












Cryocooler Controller		
Sheet: /Display/		
File: display.kicad_sch		
Title: Display		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 19/19